

Review



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Modelling behaviour in intergroup conflicts: a review of microeconomic approaches

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Microeconomic modelling offers a powerful formal toolbox for analysing the complexities of real-world intergroup relations and conflicts. One important class of models scrutinizes individuals' valuations of different group memberships, attitudes towards members of different groups and preferences for resource distribution in group contexts. A second broad class uses game theoretical methods to study strategic interactions within and between groups of individuals in contest and in conflict. After a concise discussion of some essential peculiarities of microeconomic modelling, this review provides an overview of the pertinent literatures in economics, highlights instructive examples of central model types and points out several ways forward.

This article is part of the theme issue 'Intergroup conflict across taxa'.

1. Introduction

Human intergroup conflicts can be characterized as strategic interactions between several individuals divided into two or more groups that have (at least partially) antagonistic interests. Past ages and the present abound with instances of such conflicts. They have shaped many of the boundaries of those pieces of land we call 'countries', our family trees, our institutions, our cultural memories, and maybe even some of the ways in which we think and behave [1–6].

Economic theorists try to make sense of the complexities of reality by building models, i.e. by focusing on select phenomena and describing their key components and regularities in abstract but well-defined and consistent terms [7,8]. Inevitably, these models can never capture reality in full—they always remain 'unrealistic' to some degree. Good models, one might even argue, need to remain relatively abstract in order to be broadly applicable, as overly specific models sooner or later reduce to theoretical case studies that lack generality—a horror for many theorists in economics.

The aim of this review is to trace the main approaches that modellers in microeconomics have pursued in their attempts to make sense of (select aspects of) intergroup conflicts. Completeness is not my aim—rather, I refer readers looking for exhaustive literature surveys to several such works (also see table 1). The main aim of this review, instead, is to introduce a multidisciplinary audience to what I consider the most instructive models that economics currently has in store for understanding why, how and when individual agents gang up to compete and fight with others.

The review is structured as follows. Section 2 provides a brief clarification of the terminology to be used and of some methodological peculiarities. Section 3 focuses on models of individuals' attitudes towards members of different groups and of individuals' decisions to join certain groups but not others. In §4, the focus is shifted to models of interactions within and between groups in conflict. Section 5 points out directions for future research and §6 concludes.

Table 1. Overview of relevant review papers and collected volumes.

domain	references
general	[9–12]
group-based social preferences	[13–17]
identity choice	[18–22]
contest models	[23–29]
other modelling approaches	[30–34]

2. Microeconomic modelling: a brief primer

The central mathematical framework used by microeconomic analysis is decision and game theory. Decision theory deals with contexts in which individuals' decisions do not influence other agents' outcomes and vice versa, while game theory deals with exactly such interactive contexts. (References [8,35,36] offer comprehensive introductions.)

Canonically, economists assume that agents are rational, wherein 'rationality' is defined quite undemandingly, though: in a given decision situation, an agent is rational if they consistently rank all possible outcomes of their decisions and choose to do what yields (one of) the best outcome(s). In reality, this may require that agents learn for quite some time before they can make a rational decision. Prototypical microeconomic analyses just simply skip the details of this preparatory step and look at agents' behaviour after they completed such a learning phase. A convenient way of modelling agents' rankings of possible outcomes is the use of utility functions. Agents who choose as if they were maximizing a (broad range of) well-defined utility function(s) are then even guaranteed to be making rational choices.

One merit of this modelling approach is that it forces the modeller to be explicit about their agents' relevant psychological and cognitive make, i.e. about how agents are assumed to evaluate all possible outcomes of a decision situation and about how they process the information available to them. In particular, this includes fixing which arguments enter an agents' utility function and how changes in these arguments affect utility. An example: assuming a utility function $u_i(x) = x$, where x is agent i 's eventual monetary pay-off would directly imply that (a) i exclusively cares about money and (b) cares about increases in their pay-off always to exactly the same degree, no matter how little or much money i already has. Whether and for which types of decisions this assumption captures what people really do is an empirical question and can be tested experimentally or otherwise. The challenge for behavioural and experimental economists, thus, is to find those utility functions that best approximate observed decision behaviour in a given context.

Another merit of modelling decision situations like this is that the modeller is not only forced to describe their agents' preferences (via utility functions), but also to give a complete description of the decision situation itself. Such a full description comprises four components—easy to memorize as 'PAPI'. (i) *Players*: who is making decisions and when? (ii) *Actions*: what exactly can players do and when? (iii) *Pay-offs*: what are the eventual pay-off consequences of all possible combinations of the actions of the players? and (iv) *Information*: who knows what and when? The scope of decision theory are PAPI-structures with a single decider (plus chance

events which are conveniently modelled as decisions by a pseudo-player named 'Nature' who does not receive pay-offs and thus has no strategic interests). PAPI-structures with more than one decider, accordingly, are the 'games' studied by game theory.

An important distinction that often confuses scholars untrained in game theory is the one between 'actions' and 'strategies'. An *action* is what a player does at one given point in time when it is their turn to act. A *strategy* is much more than that: a strategy is a complete plan of action for the entire game, i.e. for all points in time when the respective player could potentially act. Example: if I need to choose between actions X and Y two times and if that is my entire action space in a game, that gives me four possible strategies: XX, XY, YX and YY, meaning 'always do X', 'do X then do Y', 'do Y then do X' and 'always do Y'.

The key recipe for 'solving' decision problems in microeconomic analysis, lastly, is utility maximization. For single-player games, i.e. individual decision problems, this usually means determining the (expected) costs and (expected) benefits of all possible strategies and then applying the player's utility function to the respective outcomes. The strategy yielding highest (expected) utility then is the solution of the problem.

For games with more than one player, things are more complex, though. While the players' goal still is utility maximization, their outcomes no longer depend exclusively on their own decisions (and, possibly, chance). Game theory branches into two subdisciplines at this point: *cooperative* game theory assumes that players can form coalitions to jointly search the space of all possible combinations of their strategies and then bindingly agree to play that combination of strategies that maximizes joint utility. (Ray & Vohra [37] offers a comprehensive overview of modern coalition formation theory.) By contrast, *non-cooperative* game theory assumes that such coalitions can only be formed if they are enforced by the alignment of the involved players' interests. This is because the fundamental assumption of non-cooperative game theory is that players will maximize their individual utility, no matter what. The prime solution concept for non-cooperative games, then, is the Nash equilibrium. Such an equilibrium is a combination of strategies from which no player would want to deviate unilaterally [38].

Before we turn to looking at how the formal tools just described are applied by economists in the study of intergroup relations and conflict, some remarks are in order about key challenges of these methods themselves. Leaving aside technical pitfalls like equations that can become unsolvable or strategy spaces that can become too large to be exhaustively explored, two main modelling challenges stand out.

One challenge is that modellers may overlook important aspects of a real-world phenomenon that radically affect the strategic interaction of interest; this leads to oversimplified models. The high art of modelling, thus, requires modellers to find a balance between keeping models tractable and general by keeping them simple, while simultaneously capturing all relevant aspects of the real situation they are analysing.

The other challenge is that in games with imperfect information, i.e. when at least one player is not fully informed about at least one aspect of the game, modellers must make auxiliary assumptions about how players solve this problem. The standard approach here is to assume that players base their decisions on (usually probabilistic) beliefs about the facts that they are lacking. An equilibrium, then, not only

requires that players do not want to deviate unilaterally from their chosen strategy, but also not from (the way in which they formed) their beliefs. Adding this extra layer of beliefs and their formation complicates the analysis and opens a door for arbitrary behavioural predictions—after all, players with funny beliefs might do funny things even if being fully rational [39].

Luckily, the critical discussion of models' assumptions and empirical testing can help us overcome both challenges. Oversimplified models will not produce good predictions and wrong assumptions about beliefs can be tracked down by eliciting study participants' actual beliefs. Nonetheless, the works reviewed in the following should be evaluated against the backdrop of these challenges and thus with patient confidence in the (all too slow) workings of scientific progress via trial and error.

3. Individuals and groups: preference models and theories of identity choice

One obvious characteristic of intergroup conflicts is that the individuals involved condition their behaviour towards others on an attribute called 'group membership' that structures the global population into two or more '(social) groups'. Recently, scholars have started to ask very fundamentally how this attribute is cognitively implemented and what its primary function(s) might be [40–42]. The vast majority of works in microeconomics, however, starts from a population of agents that is already composed of distinguishable groups.

In such a population individuals then face at least two (intertwined but heuristically separable) decision problems: (i) 'how should I behave towards members of different groups?' and, in case the individual can choose their group membership, (ii) 'which group(s) should I want to belong to?' The first problem is tackled by models of group-based social preferences, the second by models of 'identity choice'.

(a) Preference models

Models of group-based social preferences appear in the economic literature most often in studies on discrimination (references [13–16] provide comprehensive overviews and reference [17] meta-analyses 77 such studies). Traditionally, economists distinguish two motives for discrimination: theories of statistical discrimination posit that deciders value everyone the same, but hold (potentially wrong) beliefs about some pay-off-relevant characteristic of members of a certain group that would make treating them fairly appear less beneficial compared to discriminating against them [43–45]. Theories of 'taste-based' discrimination, in contrast, assume that agents simply value the members of certain groups less than members of others [46]. Preference models try to describe the logic of these different valuations, to predict their behavioural consequences, and can be used to estimate the strength of different 'tastes' for discrimination.

A frequently used preference model in this context is an extension of Charness and Rabin's social preference specification [47,48]. It applies for interactions of two agents i and j . Agent i 's utility is assumed to be:

$$u_i(x_i, x_j) = (1 - w)x_i + wx_j, \text{ with } w = r(1 + Ia)\rho + s(1 + Ib)\sigma,$$

wherein x_i and x_j are player i 's and j 's pay-offs and $w \in [0, 1]$ is the weight that i puts on j 's pay-off. Full selfishness of i , e.g. would imply $w = 0$. The indicators r and s are used to discern between i being ahead of j in terms of pay-off, i.e. $r = 1$ if $x_i > x_j$ and $r = 0$ else, and i being behind j , i.e. $s = 1$ if $x_i < x_j$ and $s = 0$ otherwise. The parameters ρ and σ then measure i 's altruism and spite, respectively: $\rho > 0$ indicates a positive concern for player j when i is ahead and $\sigma < 0$ indicates that i dislikes being behind j . (When Chen & Li [48] estimated these parameters for situations without information about group memberships, e.g. they found $\rho \approx 0.43$ and $\sigma \approx -0.05$ in a student sample with $n = 536$.) When information about group memberships is provided, the indicator I in the specification becomes relevant: $I = 1$ if i and j are members of the same group and $I = 0$ otherwise. Thus, a and b capture the changes in altruism and spite, respectively, when i faces an in-group member, relative to facing an out-group member. Chen & Li [48] estimated these parameters in a large student sample ($n = 1896$) with group memberships created using a minimal group paradigm [49] and found that participants behaved more altruistically towards in-group members and more spitefully towards out-group members.

Chen and Li's results are corroborated by Müller's [50] findings, whose study represents one of the econometrically most sophisticated estimation exercises of group-based social preferences up to now and establishes that participants' discriminatory choices qualify as rational (*sensu* §2, i.e. participants' choices are compatible with behavioural predictions based on well-behaved utility functions). However, Müller also observes considerable heterogeneity of such preferences. This is in line with results by Kranton & colleagues [51,52] who point out that some participants show 'groupy' behaviour, i.e. readily discriminate against members of out-groups as soon as group memberships are made salient, while other participants appear to be consistently 'non-groupy' (about 43% in [52]). Understanding this heterogeneity in individual reactions to markers of group membership better is an important future task, both for experimenters and modellers.

At this point, it is worth noting that economists traditionally take individual preferences as fixed givens and do not often inquire their origins or possible ultimate functions [53]. Nonetheless, a handful of instructive attempts at modelling how evolutionary dynamics might result in adaptive preferences for behaviour in intergroup conflicts exists in the economic literature [54–59]. It is telling of the complexity of such models, though, that two of the six studies just cited resort to simulations in order to obtain numerical results—a workaround considered second-best by most economic theorists.

(b) Identity choice models

While preference models take group memberships as given, models of identity choice assume that individuals can decide which social groups they would like to identify with and focus on understanding the determinants of such decisions. (Comprehensive overviews are provided by [18–22].) Shayo [21], for example, specifies individual i 's utility from identifying with group J as

$$u_{iJ}(\mathbf{a}) = x_i(\mathbf{a}) - \beta_i d_{iJ}(\mathbf{a}) + \gamma_i S_J(\mathbf{a}),$$

wherein $\mathbf{a}$ is a vector describing which actions the individuals in the population are taking and $x_i(\mathbf{a})$ is i 's material pay-off

under that action profile. Identity considerations now enter the specification as follows: the individual parameters β_i and γ_i , respectively, measure the degree to which i cares about their ‘perceived distance’, $d_{ij}(\mathbf{a})$, from j ’s members and about group j ’s status, $S_j(\mathbf{a})$. ‘Non-groupy’ individuals, thus, would be characterized by $\beta_i = \gamma_i = 0$, whereas individuals with $\beta_i > 0$ care about fitting in with a group and individuals with $\gamma_i > 0$ care about group status.

Different suggestions exist in the literature about how to define the details of $d_{ij}(\mathbf{a})$ and $S_j(\mathbf{a})$. Akerlof & Kranton [60–62], for example, take $S_j(\mathbf{a})$ to be fixed ‘reputations’ of different groups and focus more on the trade-off between individual pay-off, $x_i(\mathbf{a})$, and being expected to follow certain norms and to live up to certain ideals when joining a particular group j . Their paradigmatic example is the decision which clique to join in school when different cliques impose different requirements on their members [61].

For modelling intergroup relations and conflicts, identity choice models are particularly relevant in so far as they offer a tool for describing the conditions under which certain identities (i.e. groups with specific membership compositions, norms, and ideals) are able to mobilize individually costly efforts by their members and under which conditions they are not. Bernard *et al.* [63], for example, present a model in which group status is endogenously determined by the composition of a group’s members, thus creating incentives for ‘social free-riding’. If high-status groups do not counter this effect by excluding members who do not contribute much to status, they vanish.

Sambanis & Shayo [64], furthermore, apply the identity choice approach to the study of ethnic conflicts. They assume an ethnically diverse nation state and ask: when would individuals identify with their ethnic group, thus increasing the risk of civil war between these groups, and when would they rather identify with their nation, thus decreasing the risk for internal conflict? Among other insights, their model’s analysis reveals an unfortunate vicious cycle: identification with the own ethnic group is particularly attractive for individuals with high β_i , i.e. for those with stronger ‘distastes’ for diversity in their in-group. When this leads to civil conflict between ethnic groups, the status of the nation also suffers, making it even less attractive to identify with that overarching collective.

4. Groups in conflict

In economic jargon, the two approaches discussed in the preceding §3 are ‘behavioural’, i.e. they include assumptions about individuals’ preferences that go beyond plain socially unconcerned egoism. A strength of models based on such preferences is their potential to identify behaviour that clearly deviates from egoism in strategic and non-strategic situations as well as to make motivations for such behaviour quantifiable. However, applying such an approach also runs the epistemic risk of problem shifting, in the following sense: explaining, for example, why group A fights group B by saying that the members of A hate the members of B and therefore seek B ’s destruction immediately prompts questions about the origins and stability of such spiteful preferences, and thus shifts the explanatory problem to the preference level—like said, a level that economists often simply take as given.

A separate approach in microeconomic modelling therefore exists that analyses the emergence and intensity of

conflicts between egoistic (groups of) agents, i.e. agents who have no intrinsic motivation to fight for or against anyone but rather exclusively maximize their own pay-off. The aim of these models is to understand under which structural, rather than ‘psychological’, conditions wasteful conflict appears as the most promising option choice and which factors increase or decrease the incentives for agents to engage in such behaviour. Heuristically, these models can be grouped into two classes: (i) those based on Tullock’s famous rent-seeking contest [65,66], and (ii) other types of models.

(a) Contest models

The appeal of Tullock’s contest model lies in its versatility and its ability to capture several essential features of conflicts. In its simplest form, it pits two parties against each other, say A and B , who can exert costly efforts, $x_A, x_B \geq 0$, thus increasing the probability, $P_k(x_A, x_B)$, of winning a prize $V > 0$, where $k \in \{A, B\}$ indicates the party whose perspective we are taking. If none of the parties puts any effort, i.e. for $x_A + x_B = 0$, the prize is given to either of them at equal chance, i.e. $P_k(0, 0) = 1/2$. The resulting (expected) pay-off function for party A , thus is:

$$\pi_A(x_A, x_B) = \begin{cases} V \cdot P_A(x_A, x_B) - x_A & \text{if } x_A + x_B \neq 0 \\ V/2 & \text{else.} \end{cases}$$

Note that for party B the situation is perfectly symmetric, therefore we only focus on A from now on. Two central observations at this point are: (i) efforts x_A and x_B are wasteful, i.e. any positive effort is lost and cannot be regained; and (ii) the action profile $x_A = x_B = 0$, i.e. ‘peace’, is not an equilibrium. To see (ii) consider what A should do when B does not exert effort, i.e. when $x_B = 0$. Then, A can win V with a probability larger than $1/2$ at very small cost $x_A = \varepsilon > 0$ (provided that $P_A(\varepsilon, 0) > 1/2$, which is a standard assumption). Thus, rational and egoistic parties A and B have strong incentives to deviate from the peaceful ‘solution’ of splitting the prize equally via a fair coin flip (or a similar random mechanism). In equilibrium, therefore, both parties will put in positive efforts and thus waste some resources which would not need to be wasted if A and B were able to agree on, and to stick to, the ‘peaceful solution’. For some specifications of the contest success function P_k , including Tullock’s original one, the waste of resources in equilibrium is even so large that both parties would be strictly better off under ‘peace’, rendering the equilibrium inefficient. (As a side note: it may be interesting for biologists to note that the well-known Hawk–Dove game can be construed as a special case of a Tullock contest [67].)

The literature that followed up on and extended Tullock’s original model is vast and has already been surveyed repeatedly. Extensive overviews of theoretical work are provided, e.g. by references [23–27]. References [28,29], furthermore, survey the results of empirical tests of such contest models’ predictions. Therefore, I only highlight selected dimensions along which contest models have been extended here, with particular emphasis on those relevant for modelling conflicts between groups.

When parties A and B are modelled as groups of players, several complications arise. First, models need to fix how individual efforts, say x_{ki} with i enumerating all players in group $k \in \{A, B\}$, are aggregated into a joint group effort,

technically speaking: they need to fix a ‘group impact function’ [28]. The most frequently used impact functions are summation over all efforts (called the ‘perfect-substitutes’ function), counting only the smallest effort (weakest-link), and counting only the largest effort (best-shot). Each of these impact functions has distinct effects on the incentives for players to contribute to their group’s effort. Under perfect-substitutes, for example, groups often face coordination and collective action problems. To illustrate, assume that in some equilibrium aggregate efforts are $x_A = x_B = 10$ and that group *A* has two players. Then, many combinations of individual efforts within *A* that yield $x_A = 10$ are potential equilibria, for example, fair burden sharing ($x_{A_1} = 5$, $x_{A_2} = 5$) but also free-riding by player two ($x_{A_1} = 10$, $x_{A_2} = 0$). Which of the many potential equilibria is eventually chosen by the members of group *A* often is not predictable without additional assumptions about the decision-making process within groups [68].

Second, models need to fix how the prize, if won, is distributed among the players of the winning group, technically speaking: they need a ‘sharing rule’ [69]. Again, the characteristics of this sharing rule have substantial effects on the incentives for group members to contribute individual efforts [28,70]. A sharing rule that assigns shares of the prize proportionally to individual efforts, for example, can solve the collective action problem introduced by a perfect-substitutes impact function [71]. Another interesting case, furthermore, are nested contests, where the between-group contest for the prize is followed by a within-group contest for the ‘spoils’ within the victorious group [72–74].

A third channel for complications to arise is introduced when groups and/or players are modelled as heterogeneous. Players, for example, might differ in their ability to contribute, the impact of their contribution on group effort, their valuation of the prize, their risk attitudes, the information they have about other players’ actions and/or characteristics, and more. Groups might differ in size, in their impact functions and sharing rules, in the information available to them, in the impact of their aggregate efforts on the probability of contest success, and more. The mere length of these lists of possible heterogeneities and asymmetries already indicates the vastness of the space of extended contest models that can be constructed. It is no surprise, thus, that contest research is a vibrant and growing field [27,30].

(b) Other approaches

Even if only for the purpose of structuring the exposition in this review, it may be helpful to think of the contest literature as following a top-down approach, i.e. starting from an established model and extending it to capture some selected aspects of real-world behaviour or to study theoretically interesting model variants. A complementary bottom-up approach, then, is to start from interesting real-world observations and to try to make sense of them by building models that explain the patterns emerging in these data. Just like the phenomena studied, the resulting models are diverse, of course, and thus have less structural overlap with each other than contest models do. For overviews of (parts of) this literature, see references [31–33,75].

Phenomena that have received quite some attention in this domain are ethnic conflicts and civil wars [76–82], conflicts over natural resources [83–86], economic and strategic

incentives for mass killings [87–89], geographical and social network aspects of parties’ abilities to mobilize for conflicts [90,91], and the interplay of population structure and resource distribution in shaping conflicts [92–95]. A commonality of many of the models just cited is that they abstract away from individual agents’ incentives and assume, for simplicity, that groups, once formed, are able to overcome potential internal collective action problems. Another commonality is that many of them are subjected to empirical testing using detailed conflict datasets and sophisticated econometrics. Many of these models, thus, have survived first tests against relevant archive data, so that evaluations of their ecological validity are possible.

A second cluster of relevant works on conflict in microeconomics are analyses of games that are not originally built on the Tullock contest—note, though, that some of these can be reformulated such as to become variants of Tullock contests. References [30,34] contain overviews. These works include: models of conflicts that last for multiple periods, so called ‘tug-of-war’ or ‘war of attrition’ games, for example [96–98]; models that explicitly map out spatial aspects of conflicts [99], e.g. the existence of multiple battlefields [100,101]; and the so-called ‘guns-versus-butter’ models that focus on the trade-off between allocating resources to productive activities versus trying to aggressively appropriate the products of other agents’ work, references [102–105] provide instructive examples.

5. Ways forward

As is apparent from the preceding sections, the microeconomic study of group-based social preferences and of strategic behaviour in intergroup conflicts can safely be called ‘advanced’. It offers several paradigmatic modelling approaches that can be used to analyse (aspects of) such conflicts as well as countless applications of these frameworks—with the contest literature undoubtedly being the furthest developed. Nonetheless, several directions for future research are still awaiting more intensive exploration. I highlight those that I consider most relevant for interdisciplinary work on intergroup conflicts in the nearer future.

(a) Dynamic conflicts and endogenous entry

Much of the contest literature works with games that either have a single period, multiple periods of exactly the same kind, or quite rigidly defined elimination tournament schemes. In such set-ups, all players are involved in the contest right from the beginning and usually face only one opponent at a time. For many economic applications such set-ups capture the events to be analysed very well—think, for example, of auctions, R&D races or promotion tournaments. However, these set-ups do not capture some key aspects of more anarchic conflicts, like wars in humans or expansion conflicts in territorial animal species. In these, time is continuous and the parties involved can (strategically) decide, based on evaluations of the opponents’ current resource holding potential, when and whom to aggress and how intensely. Third parties not involved initially, furthermore, can often decide whether to enter, and if so in support of whom versus alone, or to stay out of an ongoing conflict.

While there are efforts towards extending the contest framework to include such dynamic aspects [97,98,106] and allowing for more endogeneity of conflict entry decisions [27,107], much work remains to be done. Promising applications of such conflict models with more fluid timing and participation, for example, would be the game theoretic analysis of the stability of dominance hierarchies [108,109] and of territorial conflicts in both humans and non-human animals [110–112].

(b) Group formation and stability in the shadow of conflict

A second aspect of real-world intergroup conflicts that is not captured fully by the existing models, yet, is the endogenous selection of participants into adversarial parties and how this process interacts with the likelihood and intensity of conflicts. Vice versa, the endogenous decision of participants to remain loyal versus to desert their groups during an ongoing conflict also has not received much attention from modellers so far.

The literature on identity choice reviewed in §3 offers expedient formal tools and promising first attempts at applying them to study such questions [64]. Combining these with the existing population-level analyses of conflict occurrence along ethnic, religious, economic and other potential fault lines, *sensu* references [92–94], as well as using the theoretical framework provided by Shayo [21] to develop a micro-foundation of those fault line models seems particularly worthwhile. Such a micro-foundation could cross-fertilize with the development of a comprehensive theory of the cognitive foundation of identity choice decisions, in the context of conflict but also more generally [40–42], and could also prove useful for modelling the fission–fusion dynamics occurring in many group-living species [113,114].

(c) Heterogeneity within and between groups

A third promising avenue for intensified research focuses on interindividual- and intergroup-heterogeneities. As indicated in §4, many types and combinations of such heterogeneities have received attention in microeconomics already, as well as in the interdisciplinary literature [2,67,115,116]. However, possibly owing to the economic tradition of taking individual preferences as given, only relatively little theoretical work in

microeconomics has tried to shed light on the origins, cognitive underpinnings and possible adaptive dynamics of preferences governing agents' behaviour in group contexts [54–59].

Auspicious questions in this domain include, for example: which adaptive dynamics (be they biologically or culturally instantiated) would lead to more versus less heterogeneity in individual concerns about group memberships? In the terms of Shayo's model: which factors shape the evolution of β_i and γ_i [21]? Similarly, which factors shape the evolution of ρ , σ , a and b in Chen & Li's specification [48]? (The extensive psychological literature on stereotype formation should be highly informative for this question.) With respect to between-group variance: groups composed of which types of individuals would fare better in which types of conflicts and against which opponents? and relatedly: conflicts between which types of groups have better chances of being resolved peacefully?

6. Conclusion

It would not do justice to the microeconomic literatures reviewed here to attempt a conclusive evaluation of which of these approaches, if any, will eventually prove more or less successful in helping us to understand the logic of real-world intergroup conflicts. Particularly not at a time like this when the literatures on identity choice and group-based social preferences are still relatively young and developing rapidly.

Instead, I strongly advise interested readers to immerse themselves into the realms of microeconomic theory in order to identify those modelling frameworks most suitable to analyse the problems they are after. To that end, table 1 compiles an overview of overviews: for each section of this review, the table lists references to survey works that provide entry points and further comprehensive guidance into the respective literatures.

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